Section 2.9 — Arithmetic of Rational Expressions

Math 103: College Algebra

Multiplying Rational Expressions

Multiply. Simplify your answer as much as possible. You may leave your answer in factored form.

1. $\dfrac{x}{y} \cdot \dfrac{3x}{2y}$
2. $\dfrac{ab}{xy} \cdot \dfrac{x}{a^3b^2}$
3. $\dfrac{6x^2+x-12}{2x^2-5x-12} \cdot \dfrac{3x^2-14x+8}{9x^2-18x+8}$
4. $\dfrac{x^2-7x+10}{x^3-3x^2+9x} \cdot (x^3+27)$
5. $\dfrac{mn+2m+4n+8}{np-3n+2p-6} \cdot \dfrac{pq+5p-3q-15}{mq+4m+4q+16}$

Dividing Rational Expressions

Dividing. Simplify your answer as much as possible. You may leave your answer in factored form.

6. $\dfrac{a^3}{b^3} \div \dfrac{a^5}{b^2}$
7. $\dfrac{8x^7g^3}{15x^2g^4} \div \left(-\dfrac{3xg^2}{4x^2g^3}\right)$
8. $\dfrac{x^2-6x-16}{3x^2-10x+8} \div \dfrac{2x^2+13x-24}{27x^3-64}$
9. $\dfrac{u^2+7u+12}{u^2+u-6} \div (u^2+u-12)$

Least Common Multiple of Monomials

Find the least common multiple of each list of monomials. (The comma separates items on a list.)

10. $r \quad , \quad t$
11. $9mn^3 \quad , \quad 27m^3n$
12. $4x^3y \quad , \quad 8x^2y^3z \quad , \quad xy^3z^5$

Least Common Multiple of Polynomials

Find the least common multiple of each list of polynomials. (The comma separates items on a list.) Give your answer in factored form.

13. $y+2 \quad , \quad y-2$
14. $6x-12 \quad , \quad 8x-16$
15. $x^2-6x+9 \quad , \quad 7x-21$
16. $b^2-2b-35 \quad , \quad b^2+7b+10$
17. $y-3 \quad , \quad y^2+2y-15 \quad , \quad y+5$
18. $z^2-4 \quad , \quad z^2-2z \quad , \quad z^3-8$

Adding and Subtracting Rational Expressions

Add or subtract, as indicated. Simplify your answer as much as possible. You may leave your answer in factored form.

19. $\dfrac{5x^2}{3z} + \dfrac{4x^2}{3z}$
20. $\dfrac{18x^2}{5xy} + \dfrac{10y^5}{5xy}$
21. $\dfrac{5}{2c} + \dfrac{1}{6c}$
22. $\dfrac{5a}{b^2} - \dfrac{3a}{b}$
23. $\dfrac{x^2+2}{x+1} + \dfrac{6-x^2}{x+1}$
24. $\dfrac{v+8}{v^2+8v+16} - \dfrac{4}{v^2+8v+16}$
25. $\dfrac{3}{y+2} + \dfrac{5}{y-4}$
26. $\dfrac{3}{t-3} + \dfrac{6}{3-t}$
27. $\dfrac{z+5}{z^2+z-12} + \dfrac{z+2}{z-3}$
28. $\dfrac{x-2}{x^2-16} - \dfrac{1}{x^2-4x}$
29. $\dfrac{4t}{t^2-9} - \dfrac{3t-2}{t^2-8t+15}$
30. $\dfrac{w-7}{w^2+4w-5} - \dfrac{w-9}{w^2+3w-10}$

Mixed Operations

Perform the indicated operations. Be sure to obey the order of operations. Simplify your answer as much as possible.

31. $\dfrac{3}{x} + \dfrac{1}{x^2+3x} \cdot \dfrac{2x}{3}$
32. $\left(\dfrac{2x-y}{2x+y} + \dfrac{2x+y}{2x-y}\right) \div \dfrac{2x-y}{2x+y}$